SmartHull optical response to plate bending

Saeed Hamad Sultan Hamad Alketbi
Rashed Rahmah Rashed Rahmah Alhantoobi
Hazza Ali Jafar Albani Alblooshi

Applied Technology Schools - Abu Dhabi - Boys
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Abstract

SmartHull investigates polymer optical fiber sensing of bending using an ESP32 and Fiber Opt Click modules. Direct fiber bending produced receiver-voltage decreases of 8.46-10.65%. In subsequent reference-channel plate tests, a 297 × 210 × 2 mm polycarbonate specimen was manually deflected by reported amounts of 10, 20 and 30 mm across approximately 228.6 mm support spacing. Five trials produced sensing/reference ratio decreases of 1.19-4.00%. The 10 mm trial recovered to within 0.06% of baseline; the repeated 20 mm trial recovered to within 0.29%. A descriptive linear fit to all five trials gives ratio decrease (%) ≈ 0.1145 × deflection (mm) + 0.0324. Finite-rotation beam calculations provide conditional centre bending-strain estimates of approximately 2,278, 4,447 and 6,414 microstrain at 10, 20 and 30 mm. The recorded response supports continued development of bending sensing; strain and voltage-to-strain mappings remain calculated, preliminary estimates.

Test methodology

The experiments used MikroElektronika Fiber Opt Click modules, polymer optical fiber and an ESP32. Both modules used 3.3 V and a common ground; D25 drove their TX inputs at command setting 132/255. After an earlier joined-output wiring issue was corrected, initial single-channel tests used the sensing analog tap on D34. In the later two-channel configuration, the reference loop connected to D34 and the sensing path to D35. Disconnecting the reference fiber reduced D34 to the approximately 142 mV readout floor while D35 remained near 1,670 mV, identifying the reference path. The configurations are reported separately.

The polycarbonate plate spanned two supports along its 297 mm direction. Reported support spacing was approximately 9 inches (228.6 mm). Unloaded centre height was 6.00 cm; loaded heights of 5.00, 4.00 and 3.00 cm correspond to 10, 20 and 30 mm deflection. Manual pressure produced the bends. Connectors were kept still where possible; operator-reported movement and possible table shaking are noted with the affected trials.

Deflection (mm) = 10 × [6.00 − loaded centre height (cm)]

Firmware output approximately ten rows per second. Duplicate timestamps were removed and stream advancement checked. Each two-channel acquisition lasted about eight seconds and summarized approximately 92 distinct rows, including the initial rolling buffer. A fresh unloaded baseline preceded each cycle. Trial summaries use medians; repeated rows within a hold are not independent trials. Local strain and applied hand force were not measured.

$$R = V_{sensing,D35} / V_{reference,D34}$$
$$q\ (\%) = 100 \times (1 - R_{bent} / R_0)$$

R_0 is the paired unloaded ratio; q is normalized receiver-voltage decrease. No readout-floor subtraction was applied. Released changes are signed relative to each cycle baseline. Receiver voltage is an optical-response indicator; these percentages are not calibrated optical-intensity losses. The historical single-channel trials use $q_V = 100 \times (1 - V/V_0)$: the 20 mm trial excludes the first five seconds and summarizes the next 20; the 30 mm trial uses the last ten seconds. They were not pooled into the newer ratio fit.

Test methodology continued

The sensing fiber crossed the plate once, turned around at one end, and returned across the plate along a second pass. These two passes form one continuous sensing optical path, not two independent sensing channels; the separate reference loop remains a different path. Routing was clarified by the operator and prototype photographs. Pass spacing, turn radius and attachment details were not measured. The measured voltage response includes the complete routed path; no factor-of-two strain or optical-loss correction is applied.

Calculated strain used finite-rotation elastica rather than the small-deflection relation. For a planar neutral-axis profile w(x), curvature and surface bending-strain magnitude are:

$$\kappa = |w''(x)| / [1 + w'(x)^2]^{3/2}$$
$$|\varepsilon_b| = (t/2)\,\kappa$$

Here t = 2 mm and L = 228.6 mm. The numerical model represents an initially straight uniform beam, symmetric central point loading, smooth supports of negligible radius, and an inextensible neutral axis able to slide over the supports. Normal support reactions include horizontal components. It excludes support friction, broad hand contact and plate effects. Reported strains represent the calculated centre bending component. Total plate strain and fiber strain were not independently measured.

Let φ be the tangent angle at a support and θ the local tangent angle, decreasing to zero at the centre. With arc length s, support reaction magnitude N and bending stiffness EI, the half-span relation is:

$$(d\theta/ds)^2 = 2(N/EI)\sin(\varphi - \theta)$$

Define the following definite integrals, each evaluated from θ = 0 to θ = φ:

$$I_x(\varphi) = \int \cos(\theta) / \sqrt{[\sin(\varphi - \theta)]}\, d\theta$$
$$I_y(\varphi) = \int \sin(\theta) / \sqrt{[\sin(\varphi - \theta)]}\, d\theta$$

For each reported centre deflection δ, solve for φ and calculate centre bending strain using:

$$I_y(\varphi) / I_x(\varphi) = 2\delta/L$$
$$\kappa_c = (2/L)\, I_x(\varphi) \sqrt{[\sin(\varphi)]}$$
$$|\varepsilon_{b,c}| = (t/L)\, I_x(\varphi) \sqrt{[\sin(\varphi)]}$$

Strain in microstrain is $\varepsilon \times 10^6$. Numerical integration and bisection were used; the endpoint singularity was removed with $\theta = \varphi(1 - u^2)$, $0 \le u \le 1$. The implementation reproduced published smooth-support normalized-load benchmarks to within 0.00003 and recovered the small-deflection limit. No assumed modulus was needed for this geometric calculation. These checks establish numerical agreement with the selected beam model. [1,2]

For 10, 20 and 30 mm deflection, this model gives centre bending-strain magnitudes of approximately 2,278, 4,447 and 6,414 microstrain, respectively. These are calculation outputs paired with the reported deflections.

The plate width-to-span ratio is approximately 0.92, and manual pressure has a finite contact area. Engineering review will assess these geometric details and the unloaded plate profile when refining the calculated strain labels.

[1] Ohtsuki, A. (1986). An Analysis of Large Deflection in a Symmetrical Three-point Bending of Beam. Bulletin of JSME, 29(253), 1988-1995. doi:10.1299/jsme1958.29.1988.

[2] Batista, M. (2014). Large Deflections of Beam Subject to Three-points Bending. arXiv:1408.3094. https://arxiv.org/pdf/1408.3094. Smooth-support benchmarks: Table 1.

Test results

The optical-path check gave approximately 504 mV connected, 142 mV disconnected and 420 mV reconnected. The first bending experiment used the fiber alone, without the polycarbonate plate.

Table 1. Direct manual fiber bends without the polycarbonate plate.

Bend label	Baseline (mV)	Bent (mV)	Decrease	Released (mV)
Steep bend 1	486.8	434.9	10.65%	487.2
Steep bend 2	486.8	445.6	8.46%	510.2
Steep bend 3	512.0	461.9	9.80%	512.6

Labels distinguish trials; bend radius was not measured and severity was not ranked. Releases 1 and 3 returned within 0.12% of baseline. Release 2 was 4.80% above baseline after a reported change in overall fiber shape.

Table 2. Plate trials and conditional centre bending-strain estimates.

Deflection	Baseline / bent (mV)	Voltage decrease	Calculated strain (με)
10 mm target	443.2 / 448.8	−1.27%	Not assigned
20 mm held	431.9 / 423.0	2.06%	≈ 4,447
30 mm reported held	440.5 / 424.5	3.63%	≈ 6,414
30-35 mm varying	445.7 / 411.3	7.73%	Not assigned

At 20 mm, voltage recovered to 431.1 mV, 0.20% below baseline. No paired release was recorded for the exact 30 mm trial. The 30-35 mm value is a final five-second median during a changing bend. The later 10 mm target trial instead increased voltage by 1.27% and released at 452.3 mV, 2.06% above baseline; its exact held height was not explicitly confirmed. This trial is retained as a separate observation at the lower target deflection.

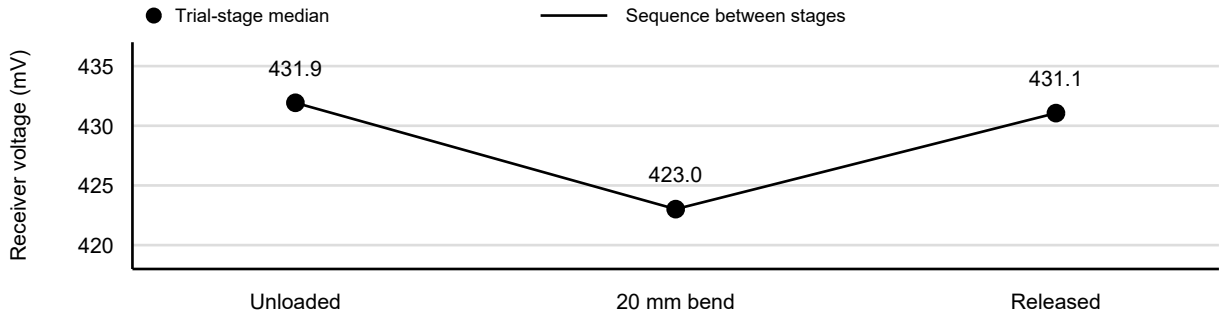


Figure 1. Medians before, during and after the 20 mm plate bend. Connecting lines indicate trial order, not a continuous time trace.

These historical single-channel values document the development of the sensing arrangement. The updated trend and voltage-to-calculated-strain mapping use the later separate-channel ratio data shown next. Earlier 900 g tests produced small voltage increases of approximately 0.54-0.58%; they are retained as exploratory load tests and are not part of the manual-deflection fit.

Test results continued

Table 3. Separate-channel plate trials; released changes are relative to the paired baseline.

Trial	Baseline ratio	Bent ratio	Ratio decrease	Released change
10 mm — Trial 1	0.79987	0.79037	1.19%	+0.06%
20 mm — Trial 1	0.78123	0.76733	1.78%	+0.74%
20 mm — Trial 2	0.78967	0.76721	2.84%	−0.29%
30 mm — Trial 1	0.78954	0.76633	2.94%	−0.86%
30 mm — Trial 2	0.78668	0.75518	4.00%	+1.35%

Movement was reported after 20 mm Trial 1; 30 mm Trial 2 was later associated with possible table shaking. Both remain displayed and are included in the descriptive all-trial fit, avoiding a fit selected for closer recovery. Their physical disturbance mechanisms were not independently verified. The 30 mm Trial 2 target was initially 40 mm, but actual deflection was corrected to 30 mm.

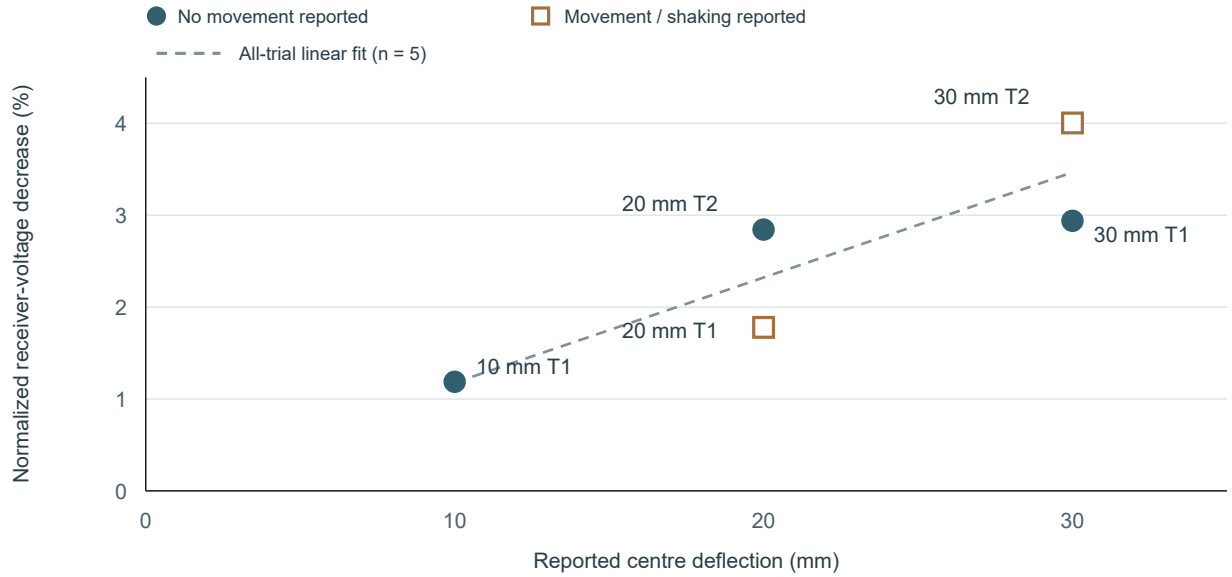


Figure 2. Normalized receiver-voltage decrease versus reported centre deflection. All five trial medians are shown. Open squares identify trials with reported movement or possible table vibration. The dashed line is an ordinary least-squares fit to all five points, drawn only across 10-30 mm.

$q\ (\%) \approx 0.11447\ \delta\ (mm) + 0.03235$

Descriptive $R^2 = 0.764$; n = 5 trial summaries.

A separate fit of q to the nonlinear model-calculated strain labels gives $q \approx 0.0005566\ \varepsilon_{calc} - 0.1209$, or $\varepsilon_{calc}\ (\mu\varepsilon) \approx (q + 0.1209) / 0.0005566$. Here $q = 100 \times (1 - R/R_0)$; this supersedes the earlier single-channel mapping. Both equations are preliminary descriptions over the sampled 10-30 mm range. R^2 describes agreement within these trials, not held-out prediction accuracy. Calculated strain is not independently measured plate or fiber strain.

The new 10 mm cycle decreased R from 0.79987 to 0.79037 and released at 0.80036. The 20 mm repeat decreased R by 2.84% and released 0.29% below baseline. At 30 mm Trial 1, the reference changed by +0.03% while sensing voltage decreased by 2.89%. These observations support a sensing-path response during bending, alongside the recorded variation between cycles.

Filtered fit and error comparison

A supplementary filtered analysis omits 20 mm Trial 1, after which movement was reported, and 30 mm Trial 2, for which possible table shaking was reported. The retained observations are 10 mm Trial 1, 20 mm Trial 2 and 30 mm Trial 1. These exclusions use operator observations; the disturbance mechanisms were not independently verified. All five measurements remain in Table 3 and Figure 2. The filtering was evaluated after inspecting the results and was not a pre-specified exclusion rule.

Analysis	Trials (n)	R^2	RMSE (percentage points)
All-trial linear fit	5	0.764	0.476
Filtered linear fit	3	0.791	0.368

Filtered equation: $q\ (\%) \approx 0.08759\ \delta\ (mm) + 0.57148$

Here δ is reported centre deflection and q is normalized receiver-voltage decrease. Both fits include an estimated intercept. $RMSE = \sqrt{[\Sigma(q_i - \hat{q_i})^2/n]}$, calculated across the trial summaries used to fit each equation. Thus the reported RMSE values describe in-sample fitting error, not independent prediction accuracy, measurement uncertainty or a confidence interval. An RMSE of 0.368 percentage points is on the scale of the percentage decrease; it is not a relative error of 0.368%.

Filtering raises descriptive R^2 from 0.764 to 0.791 and reduces fitting RMSE from 0.476 to 0.368 percentage points. The improvement is modest, and the analyses use different sets of observations. With one retained trial at each of three deflections, the filtered fit provides a preliminary sensitivity analysis rather than independent validation. The full-data fit remains the primary description of the complete recorded dataset; the filtered equation should be considered only over the sampled 10-30 mm range.

This filtering applies only to the deflection-versus-ratio analysis. The calculated-strain equation on the preceding page remains the all-five-trial fit; no filtered strain calibration has been independently established.

Future work

Phase 2 will develop the current Fiber Opt Click and ESP32 prototype into a more repeatable measurement system. The first priority is to secure the supports, plate, fiber attachment and connectors so that a defined plate deflection produces a consistent fiber shape. Mechanical stops or controlled loading will replace hand-held deflection. University laboratory review will assess the analog tap, detector response, support contact conditions and applicability of the large-deflection model. Independent strain or profile measurements will be used to check calculated strain labels.

Building on the completed separate-channel trials, longer recordings will compare sensing-only measurements with sensing-to-reference normalization under controlled source changes and steady mechanical conditions. Because the Click modules have separate emitters, reference correction will be evaluated experimentally rather than presumed to cancel source drift. The optical source will remain the Click module emitter; no laser is required for the planned prototype tests.

A larger dataset will cover several fixed deflections with repeated loading and unloading cycles. Each cycle will include unloaded readings to quantify baseline drift and recovery. Deflection, receiver voltage and any independently verified strain will be recorded together. This will allow repeatability, hysteresis, sensitivity and the usable measurement range to be assessed. The present five-trial mapping will serve as a preliminary baseline for these additional tests.

Linear regression will provide the baseline estimator. Polynomial regression, random forest and a compact multilayer perceptron will be compared where the dataset supports them. Entire cycles will be separated into training, validation and test sets. Model selection will consider held-out MAE and RMSE, inference time and ESP32 memory requirements. Deflection will be the initial prediction target; strain estimation will use mechanically verified labels.

After validation, a suitable lightweight model will be deployed on the ESP32 for experimental real-time estimates and prototype alerts. A linear model may be implemented directly; TinyML will be used only if a compact nonlinear model provides a measurable benefit. Alert thresholds will be based on the tested laboratory range. Extension to representative hull materials, multiple sensing locations and marine environmental testing is a later research objective; the present phase focuses on the laboratory plate response.

Final overview

Phase 1 established an optical-path response, recorded reversible voltage changes during direct fiber bending, and extended testing to separate sensing and reference channels at reported plate deflections of 10-30 mm. The five ratio trials show an overall positive association between deflection and normalized receiver-voltage decrease. The new 10 mm cycle and repeated 20 mm cycle recovered close to their starting ratios, while other cycles recorded shifted unloaded levels. The complete dataset is retained in the results. The linear trend and nonlinear beam calculations provide a provisional quantitative starting point for engineering review, controlled repeat testing and later ESP32 model development.